THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 8 **(b)** 7 **(c)** 11 **(d)** 9

2. 12 men can finish a job in 20 days. After 8 days, 4 more men join. The remaining work will be finished in: 1

(a) 15 days **(b)** 9 days **(c)** 12 days **(d)** 11 days

3. A price is first increased by 20% and then decreased by 20%. The net change (as percent, + for increase, − for decrease) is: 1

(a) 0% **(b)** −5% **(c)** −3% **(d)** −4%

4. A rhombus has perimeter 292 cm and one diagonal 96 cm. Its area (in cm²) is: 1

(a) 2640 **(b)** 5328 **(c)** 7008 **(d)** 5280

5. If $a : b = 3 : 5$ and $b : c = 10 : 7$ and $a + b + c = 322$, then b equals: 1

(a) 140 **(b)** 154 **(c)** 98 **(d)** 84

6. If $(x + p)(x + q) = x^2 + 7x + 12$, then $p^2 + q^2$ equals: 1

(a) 49 **(b)** 37 **(c)** 73 **(d)** 25

7. Two trains cover the same distance at speeds in the ratio 4 : 5. The ratio of the times they take is: 1

(a) 9 : 4 **(b)** 16 : 25 **(c)** 5 : 4 **(d)** 4 : 5

8. The CI on a sum for 2 years at 20% p.a. is ₹ 1100. The sum (in ₹) is: 1

(a) 2500 **(b)** 2750 **(c)** 2250 **(d)** 5000

9. A ladder 13 m long, standing in a street, rests against a house on one side when its top reaches 12 m. Without moving its foot it is turned to rest against the opposite house, reaching 5 m. The width of the street (in m) is: 1

(a) 19 **(b)** 18 **(c)** 13 **(d)** 17

10. 120 men have provisions for 30 days. After 10 days 20 men leave. The remaining provisions last the rest of the men (days): 1

(a) 26 **(b)** 36 **(c)** 24 **(d)** 20

11. The area of a square of side 25 cm exceeds that of a square of side 22 cm by: 1

(a) 75 **(b)** 9 **(c)** 484 **(d)** 141

12. If 3 : 5 = 12 : 20, then (3+5) : 5 = x : 20, where x is: 1

(a) 20 **(b)** 37 **(c)** 32 **(d)** 12

13. **Assertion (A):** A triangle with sides 6 cm, 8 cm and 10 cm is right-angled. 1

Reason (R): If the square of the longest side equals the sum of the squares of the other two sides, the triangle is right-angled.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** If 5 kg of rice costs ₹300, then 8 kg of rice costs ₹480. 1

Reason (R): The cost of rice is directly proportional to its quantity, so $\frac{\text{cost}}{\text{quantity}}$ is constant.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** $(a-b)^2 = (b-a)^2$. 1

Reason (R): $(a+b)(a-b) = a^2 - b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Evaluate $612^2 - 388^2$ using an identity. 2

17. If x : y = 7 : 3, find the ratio (x + y) : (x - y). 2

18. The cost of an item is ₹ 1500. It is marked 40% above the cost price and sold at 10% discount. Find the selling price. 2

19. Find a Pythagorean triplet in which one member is 19. 2

20. Two numbers are in the ratio $7 : 3$ and their difference is 52. Find the numbers. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. The value of a car depreciates by 10% every year. If its present value is ₹ 500000, find its value after 3 years. 3

22. A rhombus has perimeter 292 cm and one diagonal 96 cm. Find the other diagonal and the area. 3

23. A mixture contains milk and water in the ratio 3:2. When 10 L of water is added, the ratio of milk to water becomes 3:4. Find the quantity of milk and water in the original mixture. 3

24. If $a + b = 10$ and $ab = 24$, find $a^2 + b^2$ and $(a - b)^2$. 3

OR

Divide ₹ 2600 into three parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. 3

25. A shopkeeper buys an article at ₹ 800, marks it up by 50% and gives a discount of 20%. Find his gain or loss percent. 3

OR

Two ships leave a port, one sailing east at 8 km/h and the other south at 15 km/h. How far apart are they after 3 hours? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Sunita in Amritsar mixes curd and water in the ratio $3 : 2$ to make 10 litres of lassi. 5

(i) What fraction of the lassi is curd?

(ii) How many litres of curd are used?

(iii) How many litres of water are used?

(iv) She wants to make 25 litres of lassi in the same ratio. Find the quantities of curd and water and show that the ratio is unchanged.

27. A vessel holds 60 L of a mixture of milk and water in the ratio 2 : 1. 15 L of the mixture is removed and replaced by 15 L of water. Find the new ratio of milk to water. 5

OR

A shopkeeper marks an article 50% above its cost price. During a sale he gives a discount of 20% and still earns a profit of ₹ 240. Find the cost price. 5

END OF QUESTION PAPER — Total: 50 Marks

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